Augustin's Reference Guide to Mathematical Set Theory in LaTeX

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A comprehensive reference guide for formatting mathematical sets, operations, quantifiers, and intervals in LaTeX. Curated and maintained by **Augustin Thomas**.

All commands below must be wrapped in math mode delimiters (`$` or `\[ \]`).

Quick Reference

1. Common Number Sets (Blackboard Bold)

Note: To use these symbols, you must include `\usepackage{amssymb}` or `\usepackage{amsfonts}` in your document's preamble.

Set Name	LaTeX Code	Rendered Output
Natural Numbers	`\mathbb{N}`	$\mathbb{N}$
Integers	`\mathbb{Z}`	$\mathbb{Z}$
Rational Numbers	`\mathbb{Q}`	$\mathbb{Q}$
Real Numbers	`\mathbb{R}`	$\mathbb{R}$
Complex Numbers	`\mathbb{C}`	$\mathbb{C}$
Empty Set	`\emptyset` or `\varnothing`	$\emptyset$ or $\varnothing$

2. Core Set Operations

Operation	LaTeX Code	Rendered Output
Union	`A \cup B`	$A \cup B$
Intersection	`A \cap B`	$A \cap B$
Set Difference / Minus	`A \setminus B`	$A \setminus B$
Cartesian Product	`A \times B`	$A \times B$
Symmetric Difference	`A \triangle B`	$A \triangle B$
Complement	`A^c` or `A'`	A^c or A'

3. Relations and Membership

Meaning	LaTeX Code	Rendered Output
Element of / Belongs to	`x \in A`	$x \in A$
Not an element of	`x \notin A`	$x \notin A$
Subset of (or equal to)	`A \subseteq B`	$A \subseteq B$
Strict / Proper Subset	`A \subset B` or `A \subsetneq B`	$A \subset B$ or $A \subsetneq B$
Superset of	`A \supseteq B`	$A \supseteq B$
Not a subset of	`A \nsubseteq B`	$A \nsubseteq B$

4. Quantifiers and Logic

Meaning	LaTeX Code	Rendered Output
For all / For each	`\forall`	$\forall$
There exists	`\exists`	$\exists$
There exists a unique	`\exists!`	$\exists!$
Does not exist	`\nexists`	$\nexists$
Implies	`\implies`	$\implies$
If and only if (Equivalent)	`\iff`	$\iff$
Therefore	`\therefore`	$\therefore$
Because / Since	`\because`	$\because$

5. Intervals and Bounds

Interval Type	LaTeX Code	Rendered Output
Closed Interval	`[a, b]`	$[a, b]$
Open Interval	`(a, b)`	(a, b)
Left-Open, Right-Closed	`(a, b]`	$(a, b]$
Left-Closed, Right-Open	`[a, b)`	$[a, b)$
Infinity	`\infty`	∞
Negative Infinity	`-\infty`	$-\infty$

6. Power Sets, Collections, and Families

When denoting families of sets, collections, topologies, or power sets, it is standard practice to use a calligraphic font. You can achieve this using the `\mathcal{}` command.

Concept	LaTeX Code	Rendered Output
Power Set	`\mathcal{P}(A)` or `2^A`	$\mathcal{P}(A)$ or 2^A
Family of Sets	`\mathcal{F}`	$\mathcal{F}$
Collection of Sets	`\mathcal{C}`	$\mathcal{C}$
Topology	`\mathcal{T}`	$\mathcal{T}$

Large Indexed Operations

Indexed Union

- **Inline Mode (`$...$`):** `$\bigcup_{i=1}^{n} A_i$` → $\bigcup_{i=1}^{n} A_i$
- **Display Mode (`\[...\]`):** `\[ \bigcup_{i=1}^{n} A_i \]` →

$$\bigcup_{i=1}^{n} A_i$$

Indexed Intersection

- **Inline Mode (`$...$`):** `$\bigcap_{i=1}^{n} A_i$` → $\bigcap_{i=1}^{n} A_i$
- **Display Mode (`\[...\]`):** `\[ \bigcap_{i=1}^{n} A_i \]` →

$$\bigcap_{i=1}^{n} A_i$$

Set-Builder Notation & Braces

Because curly braces `{ }` are used for grouping code in LaTeX, you must escape them with a backslash `\{ \}` to display them visually.

- **Standard Notation:**

  ```
  \{ x \in \mathbb{R} \mid x > 0 \}
  ```

  Output: $\{x \in \mathbb{R} \mid x > 0\}$

- **Auto-Scaling Notation (For tall elements like fractions):**

  ```
  \left\{ x \in \mathbb{Q} \;\middle\vert\; x = \frac{1}{n} \right\}
  ```

  Output: $\left\{ x \in \mathbb{Q} \;\middle\vert\; x = \frac{1}{n} \right\}$

Formatting Tips

1. Text Inside Math Mode

Use `\text{...}` (requires `\usepackage{amsmath}`) to add normal words inside your equations so they do not render in cramped math italics:

```
\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \quad \text{if } x < y
```

Output: $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \quad \text{if } x < y$

2. Fine-Tuning Math Spacing

If your symbols look too crowded, inject small spaces using these spacer commands:

- `\,` (thin space)
- `\;` (medium space)
- `\quad` (large space equal to the width of one character)

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